

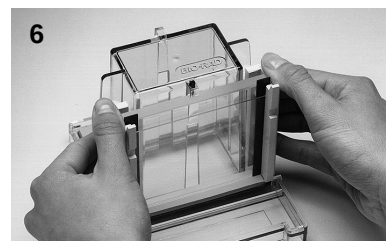
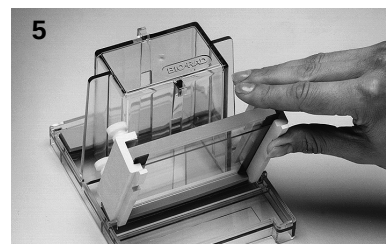
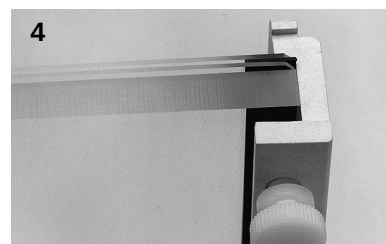
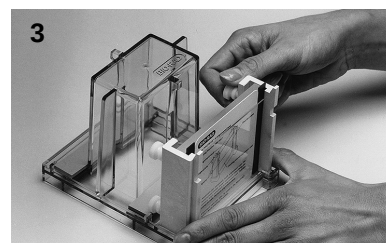
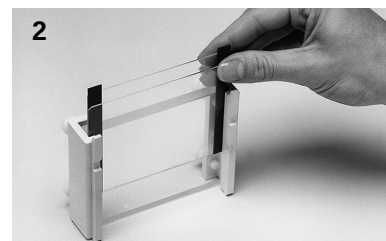
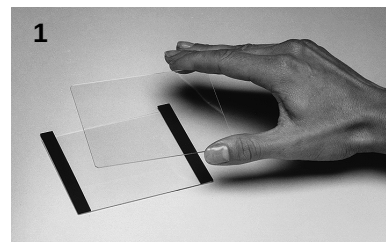
Mini-PROTEAN® II Cell Assembly Guide

Assembling the Glass Plate Sandwich

1. Assemble the gel sandwich on a clean surface. Lay the longer rectangular glass plate down first. Place two spacers of equal thickness along the short edges of this plate. Next, place the shorter glass plate on top of the spacers so that the bottom ends of the spacers and glass plates are aligned. At this point, the spacers should be sticking up above the longer glass plate approximately 5 mm.
2. Loosen the four screws on the clamp assembly and stand it up so that the screws are facing away from you. Firmly grasp the glass plate sandwich and gently slide it into the clamp assembly with the longer glass plate against the front face of the acrylic pressure plate. Tighten the top screws of the clamp assembly.
3. Place the clamp assembly into the alignment position on the casting stand so that the clamp screws face toward the casting positions. Loosen the top two screws to allow the plates and spacers to sit flush against the base. Insert an alignment card between the glass plates to keep the spacers in the proper upright position. Hold the clamps at the base of the stand and re-tighten the top two clamp screws.

Note: Failure to use the alignment position to properly orient the glass plates can result in casting leaks while pouring the gel, or buffer leaks during the run.

4. Remove the clamp assembly from the alignment position and tighten the bottom two clamp screws. Check for proper alignment by inverting the gel sandwich and looking at the surface of the two glass plates and spacers. All three surfaces should be flush.
5. Transfer the clamp assembly to one of the casting positions on the casting stand. If two gels are to be cast, place the clamp assembly on the side opposite the alignment slot to make aligning the next sandwich easier. To attach the clamp assembly, butt the acrylic pressure plate against the wall of the casting position at the bottom, so that the glass plates rest on the rubber gasket.
6. Snap the acrylic plate underneath the overhang of the casting position by pushing with the white portions of the clamps. Do not push against the glass plates or spacers, as this could break the plates. The gel sandwiches are now ready for casting.



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Assembling the Upper Buffer Chamber

1. To insure a leakproof seal, make sure that the gray, U-shaped, inner core gaskets are clean and that they are installed with the notched side exposed for contact with the gel sandwich.
2. Lay the electrode assembly on a flat surface. With the glass plates facing the inner core (and the clamp screws facing up), place the top of the clamp assembly underneath the locator slots on the electrode assembly and rest the bottom of the clamp assembly on the base plate. Carefully slide the clamp assembly forward, until the inner glass plate of the gel sandwich butts against the notch in the U-shaped gasket.

Note: If the clamp is introduced at too steep an angle, the inner glass plate will pinch the notched gasket and leaking will result. Lubricate the raised portions of the gasket with a drop of running buffer or water to help the glass plate sandwich slide in properly. It is extremely important that the short glass plate is firmly seated against the notch in the gasket, or buffer leaks will occur. **Visually inspect the glass plate/notch interface to insure correct assembly.**

3. When the inner plate is in position, push **down** on the base of the clamp assembly until the latch engages each side.
4. Turn over the electrode assembly and attach another clamp assembly to the other side in the same manner.
5. If only one gel is to be run, a buffer dam may be made by inserting a short (inner) glass plate and a long (outer) glass plate without spacers into the clamp assembly. Align the plates using the alignment position of the casting stand, tighten the clamp screws, and snap onto the electrode assembly as usual.

